

A Centralized Multi-Agent DRL-Based Trajectory Control Strategy for Unmanned Aerial Vehicle-Enabled Wireless Communications

GETANEH BERIE TAREKEGN ^{1,2} (Member, IEEE), RONG-TERNG JUANG ³ (Member, IEEE),
BELAYNEH ABEBE TESFAW ⁴, HSIN-PIAO LIN ³ (Member, IEEE), HUAN-CHIA HSU²,
ROBEL BERIE TAREKEGN ⁵, AND LI-CHIA TAI ^{1,2} (Senior Member, IEEE)

¹Department of Electrical and Computer Engineering, National Yang Ming Chiao Tung University, Hsinchu 30010, Taiwan

²Institute of Electrical and Control Engineering, National Yang Ming Chiao Tung University, Hsinchu 30010, Taiwan

³Institute of Aerospace and System Engineering, National Taipei University of Technology, Taipei 10608, Taiwan

⁴Department of Electrical Engineering and Computer Science, National Taipei University of Technology, Taipei 10608, Taiwan

⁵Department of Electrical Engineering and Computer Science, National Yang Ming Chiao Tung University, Hsinchu 30010, Taiwan

CORRESPONDING AUTHOR: LI-CHIA TAI (e-mail: j.tai@nycu.edu.tw).

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ABSTRACT Unmanned aerial vehicles (UAVs) are becoming increasingly popular as mobile base stations due to their flexible deployment and low-cost features, particularly for emergency communications, traffic offloading, and terrestrial communications infrastructure failures. This paper presents an autonomous trajectory control method for multiple UAVs equipped with base stations for UAV-enabled wireless communications. The objective of this work is to address the optimization challenge of maximizing both communication coverage and network throughput for ground users. The proposed multi-aerial base station trajectory control (MATC) scheme employs a two-stage learning approach. Initially, we developed a long short-term memory-based link quality estimation model to assess each user's link quality over time. The trajectory of the aerial base stations is then continuously adjusted through a centralized multi-agent deep reinforcement learning algorithm to optimize communication performance. We evaluated our proposed system using real channel measurement data, i.e., amplitude and phase signal information. Notably, the proposed approach operates solely on received signals from users, without requiring knowledge of their specific locations. The proposed MATC strategy achieves 97.41% communication coverage while maintaining satisfactory system throughput performance. Numerical results demonstrate that the proposed method significantly enhances both communication coverage and network throughput in comparison to the base line algorithms.

INDEX TERMS Fair communications, long short-term memory, multi-agent DRL, trajectory control, UAV communications.

I. INTRODUCTION

The swift expansion of the Internet of Things (IoT) and emerging applications such as augmented reality, virtual reality, face recognition, and online gaming have generated a compelling demand for seamless networks and on-demand services [1], [2]. This presents a critical challenge to conventional cellular

networks. Furthermore, upcoming IoT applications, such as localization, tracking objects, video processing, and vehicle-to-vehicle communications, require efficient and widespread high-speed networks [3], [4]. Notably, unmanned aerial vehicles (UAVs), commonly known as drones, have recently emerged as a promising solution for delivering seamless

wireless connectivity, complementing the existing terrestrial communication infrastructure [5]. Various scenarios involving UAVs equipped with base stations (BSs) have been explored. Aerial base stations (ABS) can substitute for and assist fixed BSs, enhancing communication services to IoT devices because of their agility, adaptable altitude, and mobility features [6]. Consequently, they can deliver high-quality and on-demand services in the event of ground BS failure, emergency communications, and traffic offloading. However, careful design of air-to-ground channel modeling and UAV trajectory control is essential for UAV wireless networks to ensure long-term effective network connectivity and reliable coverage.

A. RELATED LITERATURE AND MOTIVATIONS

The work [7] and [8] focused on extracting the air-to-ground channel characteristics by utilizing machine learning approaches in ABS-enabled wireless communications. For example, in [7], a k -means clustering algorithm accurately gauged ground user link quality, while CNN was employed to assist in user selection for ABS networks. In both [7] and [8], the received signal strength of ground users was used to analyze the characteristics of the air-to-ground 3D wireless channel. Meanwhile, early research contributions addressed the UAV trajectory control problem through conventional optimization-based approaches [9], [10], [11], [12], [13], and shallow learning methods [14], [15], [16]. Due to numerous environmental variables and unpredictable system behavior (such as ground user behavior and channel status), making sequential decisions mathematically in real time can be challenging [17], [18]. The result is inaccurate modeling and reduced system performance in terms of communication coverage and network throughput.

In contrast, artificial intelligence has emerged as a powerful force in optimizing and decision-making processes, particularly in handling complex and dynamic environments [19]. Integrating artificial intelligence into ABSs holds promise for enhancing their communication and networking capabilities, ultimately improving IoT services. Reinforcement learning (RL), in particular, has garnered increasing attention for ABS networks, encompassing tasks such as trajectory planning, resource allocation, and energy optimization [20], [21]. RL facilitates autonomous decision-making for ABS in dynamic and complex environments, thereby enhancing network communications performance [19], [21].

Furthermore, deep reinforcement learning (DRL) combines reinforcement learning with deep neural networks (DNNs) to infuse intelligence and enhance learning speed, especially in addressing large-scale problems with numerous states and actions [17], [18], [19]. In comparison to traditional optimization-based techniques, reinforcement learning specifically the DRL algorithm provides the following advantages: 1) *scalability*- DRL is more efficient at handling large-scale problems with high-dimensional environmental variables (i.e., state and action spaces) [22]. However, classical optimization methods cannot handle continuous state and action spaces;

2) *adaptability*- DRL-based solutions effectively resolve real-time dynamic decision-making problems in an unpredictable and time-varying network environment, particularly when user behavior changes over time. However, it is challenging to perform sequential decision-making problems mathematically in real-time applications [18], [19], [20], [21], [22], [23]. Additionally, classical optimization algorithms have low solution efficiency and are unable to adapt to the dynamic nature of the environment for solving ABS-enabled wireless networks; and 3) *learning strategy*- The DRL can discover optimal policies by interacting directly with a target environment to adapt to an unknown wireless environment. However, traditional optimization methods require predefined models to construct trajectory control policies. Furthermore, it often involves solving complex mathematical models that are computationally intensive and time-consuming, making them less suitable for real-time applications [23]. A DRL-based solution, on the other hand, has the ability to learn optimal strategies automatically from past experiences and is more adaptable and generalizable.

Recent research has showcased the effectiveness of both model-free conventional RL [24], [25], [26] and DRL [27], [28], [29] in controlling ABS trajectories to maximize system rewards, addressing the previously mentioned challenges. Notably, these methods have typically focused on scenarios involving a single UAV deployment with a horizontal trajectory [12], [30], a fixed number of users [16], or a static multi-UAV deployment [16]. Large-scale environments (e.g., large campuses, cities) with a nonfixed number of users require multiple ABS to operate cooperatively to satisfy users' needs, which is the main focus of this article. In addition, most prior works [25], [27], [29], and [31] have considered a single agent for multi-UAV-enabled wireless communications. In our proposed framework, we extend the principles of single-agent DRL to address the complexities inherent in scenarios involving multiple ABS and to help each ABS make independent decisions. The work in [32] utilized a decentralized learning scheme to construct an ABS trajectory control strategy. Since each ABS has limited information, achieving global optimal solutions is difficult, resulting in lower accuracy and system performance. Here, our proposed approach adopts a centralized learning framework with decentralized execution. During the execution phase, each ABS independently adjusts its deployment position based on the information it observes in the dynamic environment.

B. CONTRIBUTIONS

In this article, we address a wireless communication network featuring multiple ABSs and ground users. We present a centralized multi-agent DRL-based multi-ABS trajectory control (MATC) scheme, aimed at ensuring prolonged reliable coverage and effective network connectivity. Our proposed framework extends the principles of single-agent DRL to adeptly address the complexities inherent in scenarios involving multiple agents [20], [33]. Each ABS assumes the role

of a learning agent, interacting within the environment and collaboratively learning optimal mobility control policies through a centralized component.

In light of the works mentioned above, the proposed method has the following novelty and contributions.

- 1) We investigate the 3D trajectory control strategy of multiple ABSs in a UAV-enabled wireless communications system rather than single UAV scenario found in previous studies [10], [12], [28], and [30]. We formulate the optimization problem to maximize the coverage score and ensure satisfactory system throughput for ground users by setting a pre-defined threshold value. In practice, network operators usually do not have access to user location information due to privacy concerns. Accordingly, the proposed method operates in continuous control of the UAV trajectory task based on received signals from users without any prior knowledge of their specific locations, distinguishing it from methodologies outlined in [9], [10], [11], [12], [13], [14], [15], [16], [24], [25], [26], [27], [30].
- 2) Optimally solving the formulated problem using conventional optimization techniques [9], [10], [11], [12], [13] poses challenges due to the coupling between optimization variables and the nonlinearity of the underlying models, specifically the link quality estimation model and UAV trajectory control policy. To address this issue, we model dynamic trajectory control of ABSs as a Markov game by defining the system state, the action space and the reward function. Here, we convert the transformed problem into a multi-agent extension of the Markov decision process (MDP) by letting each ABS act as an agent. Then a centralized multi-agent DRL algorithm is proposed to generate global ABS trajectories, ensuring long-term reliable coverage and effective network connectivity. Additionally, a long short-term memory (LSTM)-based link quality estimation model is employed to assess the link quality of each ground user over time by estimating its modulation and coding scheme (MCS) index.
- 3) The performance of the proposed approach is evaluated using real-world data rather than theoretical channel modeling to demonstrate its effectiveness in ABS-enabled wireless communications. A comparison between the proposed MATC scheme and the baseline algorithms is also provided to demonstrate the validity of the proposed approach.

The subsequent sections of this article are organized as follows. Section II presents the system model and formulation of the MATC problem. Section III provides a detailed presentation of the proposed centralized multi-agent DRL framework. Section IV presents numerical results based on real-world user channel data to assess the effectiveness of the proposed solution. Finally, Section V offers concluding remarks and discusses potential avenues for future work.

TABLE 1. Experimental Setup Parameters of Air-to-Ground Signal Measurements of the ABS-Enabled Wireless Communication System

Parameter	Value
Number of ABSs	3
Channel Bandwidth	3.2 MHz
Transmitter Power	33 dBm
Modulation	OFDM/QPSK
Carrier Frequency	860 MHz
Antenna	Dipole
FFT Size	2048
Pilot Subcarrier	273
Subcarrier Δf	1.5 kHz
Guard Band Subcarrier	143

II. SYSTEM MODEL AND PROBLEM FORMULATION

This section presents the multi-agent system model and then formulates the problem for the multi-ABS trajectory control strategy.

A. SYSTEM MODEL

As depicted in Fig. 1, we consider ABS-enabled down-link wireless communication system, comprising B ABSs to serve U ground users. ABSs are linked to the core network through a ground gateway. The set of ABSs is denoted as $\mathbf{B} = \{b_1, b_2, \dots, b_B\}$, and the set of users is denoted as $\mathbf{U} = \{u_1, u_2, \dots, u_U\}$. Additionally, $\mu_b[t]$ denotes the set of users served by the b -th ABS at time slot t . To ensure stable wireless communication and high data rates, the ABS hovers at different altitudes, ranging from a minimum $h_{\min}$ to a maximum $h_{\max}$. The position of the b -th ABS at time slot t is denoted as $L_b[t] = \{x_b[t], y_b[t], h_b[t]\}$, and the distance between the i -th and j -th ABSs can be expressed as $d_{j,i}[t] = \|L_i[t] - L_j[t]\|$. This distance must exceed a threshold $d_{\min}$ to prevent collisions.

In this study, practical experiments were conducted within the vicinity of the National Taipei University of Technology campus and surrounding roads to acquire real-world user channel data. Typical wireless communication systems operate within the 860 MHz to 900 MHz carrier frequency range to provide better data rates, particularly for applications like Internet of Things devices, industrial monitoring, and vehicle-to-vehicle communications [7], [34]. For the experiments, 4.92 million user channel samples were collected. The radio signal employed in this research is based on orthogonal frequency-division multiplexing (OFDM). The OFDM signal was transmitted at a frequency of 860 MHz with a bandwidth of 3.2 MHz. Additionally, both the transmitter and receiver used a dipole antenna. The conversion of OFDM baseband signals to radio frequency signals was facilitated using a universal software radio peripheral B210 from National Instruments. The implemented software defined radio system considers path loss, slow fading, and fast fading. The parameters utilized for generating the OFDM signal are outlined in Table 1.

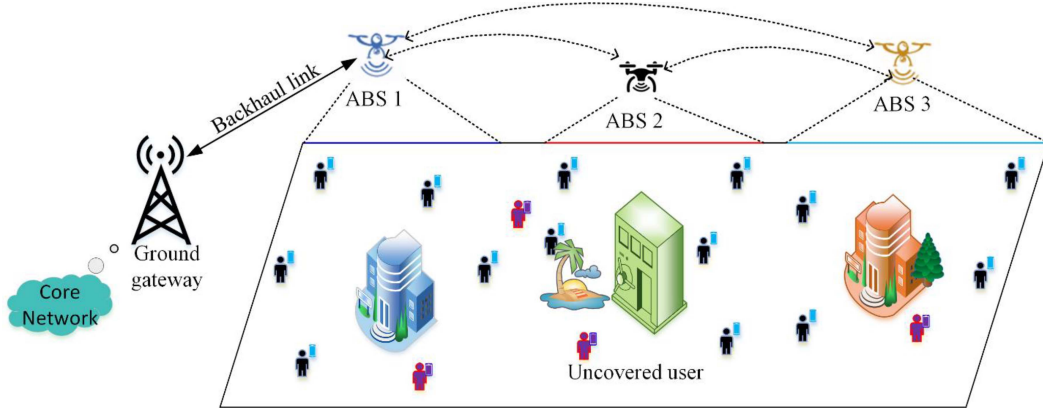


FIGURE 1. Considered multi-ABS-enabled wireless communication system model. The blue-black and red ground users represent the covered and uncovered users in the deployed ABS due to blockage and suboptimal ABS deployment, respectively.

To synchronize their locations with the received user channels during data collection, the ABSs were equipped with global positioning systems at each time slot. All ABSs were allocated orthogonal spectrum resources with identical total bandwidth to prevent interference. In realistic scenarios, users exhibit random movement within the target area. The study [35], [36] modeled user mobility using the random walk model, which accurately represents real-time user behavior. Consequently, user channel data was collected under two categories of user mobility: slow speed (< 1.4 m/s) and fast speed walking (between 1.4 m/s and 2.5 m/s). The flight altitude of the aerial base station was constrained between 30 and 50 meters to provide an appropriate line-of-sight from obstacles, buildings, and terrain features.

B. PROBLEM FORMULATION

The objective of the MATC problem is to maximize both communication coverage and the attainable user data rate by selecting suitable trajectories for each ABS across time slots. Each ground user is connected to one ABS with a strong signal at each time slot to ensure a stable and robust connection. The status of the link between the b -th ABS and the u -th user at time slot t is denoted as $\chi_{b,u}[t] \in \{0, 1\}$, where unity signifies that the user is served by the ABS, and zero indicates otherwise. When the received user data rate exceeds a threshold value, the communication association status between ABS b and the ground IoT device is set to 1. Subsequently, the communication coverage score is defined as:

$$C[t] = \frac{1}{U} \sum_{b=1}^B \sum_{u=1}^U \chi_{b,u}[t] \quad (1)$$

Similarly, we defined the data rate of the u -th user served by the b -th ABS at time slot t as $\Pi_{b,u}[t]$. The total data rate is given by

$$\Pi[t] = \sum_{b=1}^B \sum_{u=1}^U \chi_{b,u}[t] \times \Pi_{b,u}[t] \quad (2)$$

Please note that $\Pi_{b,u}[t]$ is a function dependent on the MCS used in the link (i.e., $\Pi_{b,u}[t] = f(m_{b,u}[t], b \in \mathbf{B}, u \in \mathbf{U})$). Denote the MCS index for the link between the b -th ABS and the u -th user at time slot t as $m_{b,u}[t]$, with the index required to be greater than a threshold $m_{\min}$ to ensure quality of service. As indicated in [37], the signal-to-interference-plus-noise ratio (SINR) may be adequate for regulating channel quality in terrestrial base stations due to their static nature. However, ABSs, being dynamic, pose challenges in identifying air-to-ground channel characteristics solely based on the SINR parameter. Therefore, this paper adopts amplitude and phase signal information to assess the link quality of each user. Consequently, mapping the received amplitude and phase information to the MCS index becomes essential for continuously controlling network performance along the ABS trajectory.

The amplitude information is typically associated with the modulation order, whereas the phase information is linked to the coding rate. As a result, amplitude information is labeled with the modulation order using signal-to-noise ratio (SNR) values, while the phase signal information is clustered into the coding rate using the k -means algorithm. Consequently, the MCS indices of the users can be mapped based on their labeled modulation order and coding rate. Denoting the positions of all the ABSs in a vector as $\mathbf{L}[t] = \{L_1[t], L_2[t], \dots, L_B[t]\}$, the MATC problem can be expressed mathematically as follows.

$$\begin{aligned} \hat{\mathbf{L}}[t] = & \arg \max_{\mathbf{L}[t]} C[t] \times \Pi[t] \\ \text{s.t. } & c1 : \mu_i[t] \cap \mu_j[t] = \emptyset, i \neq j \\ & c2 : m_{b,u}[t] \geq m_{\min} \\ & c3 : d_{i,j}[t] \geq d_{\min} \\ & c4 : h_{\min} \leq h_b[t] \leq h_{\max} \\ & c5 : 0 \leq v_u[t] \leq v_{\max} \end{aligned} \quad (3)$$

where $h_b[t]$ and $v_u[t]$ are the height of the b -th ABS and the velocity of the u -th user, respectively, at time slot t . Here,

constraint $c1$ is in place to ensure that one ABS does not overlap with another, thereby enhancing ABS utilization efficiency. Constraint $c2$ imposes minimum MCS index constraints on each user to ensure service quality. The MCS index of each user is selected from the provided MCS set (refer to [38, p. 2], Table 1 for details). Constraint $c3$ represents the minimum distance between two ABSs, while constraints $c4$ and $c5$ specify the altitude boundaries for the ABSs and the speed limitations for the users, respectively.

III. PROPOSED MATC SCHEME

This section describes how to transform the MATC problem into a learning task by defining DRL components that learn how to interact with the environment. The training process aims to maximize expected long-term rewards.

A. PROBLEM TRANSFORMATION WITH DRL

The MATC problem can be defined as a multi-agent MDP due to the involvement of multiple ABS. The proposed MATC policy employs several agents to train and govern ABS movements, aiming to maximize system reward by concurrently interacting within a shared environment and executing joint actions. The repositioning of each ABS can be dynamically adjusted to enhance communication services for ground users. In this paper, we leverage DRL algorithm to solve the MATC problem. The advent of DRL marks a significant advancement by combining reinforcement learning with deep neural networks [20]. This integration accelerates the learning process and enhances RL performance, particularly in large scale problems characterized by numerous states and actions.

In uncertain and stochastic environments, decision-making problems can be represented as MDPs [19], [33]. To address the MATC problem, this study formulates the time-sequential decision process as an MDP and employs a multi-agent DRL approach. In cases where an MDP is stationary, the controller relies solely on the current state, disregarding the entire historical context of the operation. Ergodicity in an MDP ensures that the transition space can be thoroughly explored, enabling an agent to construct an accurate policy after a sufficient number of cycles. Therefore, this work adopts an ergodic MDP to converge the proposed centralized multi-agent DRL-based MATC strategy.

Multi-agent MDPs are typically defined as a quintuple (F, S, A, P, R) , where F is the set encompassing all agents indexed from 1 to B , S represents the state space, A denotes the collective action spaces for all agents, P signifies the probability of state transition within the system, and R is the joint reward function applicable to any given agent. The objective of this paper is to determine an optimal policy, denoted as π^* , that simultaneously maximizes both communication coverage and user throughput in the context of the MATC problem. The MDP governing a UAV trajectory control agent is comprised of three fundamental elements: the state space, the action space, and the reward function.

1) ACTION SPACE

The action executed by the b -th ABS at time slot t is denoted as $a_b[t]$. In the context of the DRL system, the ABS's action corresponds to its flying direction. Each ABS possesses the capability to navigate in discrete directions, with nine predefined actions constituting the set {left, right, up, down, left-up, right-up, left-down, right-down, no movement}. Even if the actions are discrete, the movement of the ABS is still continuous because it follows the physics of aerial dynamics. Following the selection of a particular action, the ABS moves one meter in the specified direction. All DRL agents have the same action space.

2) STATE SPACE

The state of the b -th ABS at time slot t is represented as $s_b[t] = (L_b[t], \mu_b[t], m_{b,u}[t])$, where $L_b[t]$ denotes the location of the ABS, $\mu_b[t]$ signifies the count of served users, and $m_{b,u}[t]$ corresponds to the MCS index of the served users. An LSTM-based LQE model is utilized to determine the link quality of users at each time slot.

3) REWARD FUNCTION

The actions performed by the agents in the target environment are geared towards maximizing the overall system reward. To achieve this objective, the reward function is designed to enhance both communication coverage ($C[t]$) and system throughput ($\Pi[t]$), as outlined

$$r[t] = C[t] \times \Pi[t] - \sum_{i=1}^3 \alpha_i[t] \times p_i \quad (4)$$

Notably, $\alpha_1[t] \in \{0, 1\}$ is the penalty coefficient of $C[t]$ regardless of whether the reward function is satisfied; $\alpha_2[t] \in \{0, 1\}$ is the penalty coefficient of $\Pi[t]$ regardless of whether the reward function is satisfied; $\alpha_3[t]$ signifies whether or not the penalty coefficient of the ABS flight or constraint $c3$ is satisfied or not; p_1 denotes the penalty value of $C[t]$, which is a negative reward; p_2 denotes the penalty value of $\Pi[t]$, which is a negative reward; If an agent flies beyond the target area or violates $c3$, then p_3 cancels the corresponding action of the ABS and updates $s_b[t + 1]$ accordingly.

B. TRAINING PROCESS OF THE PROPOSED MATC SCHEME

To develop a continuous MATC strategy, we applied a combination of LSTM and multi-agent DRL algorithms. The LSTM link quality estimation model was trained using real-world user channel data. A total of 4.92 million user channel samples were collected for the experiments. The proposed model was trained by inputting preprocessed channel information into the LSTM algorithm, with the corresponding link quality level (i.e., MCS index) serving as the target value. In the experiment, 80% of the data (approximately 3.94 million samples) were randomly selected for training the channel estimation model, while the remaining 20% were reserved for evaluation purposes. Employing the trial-and-error method [39], hyperparameters of the LSTM network, such as the number of

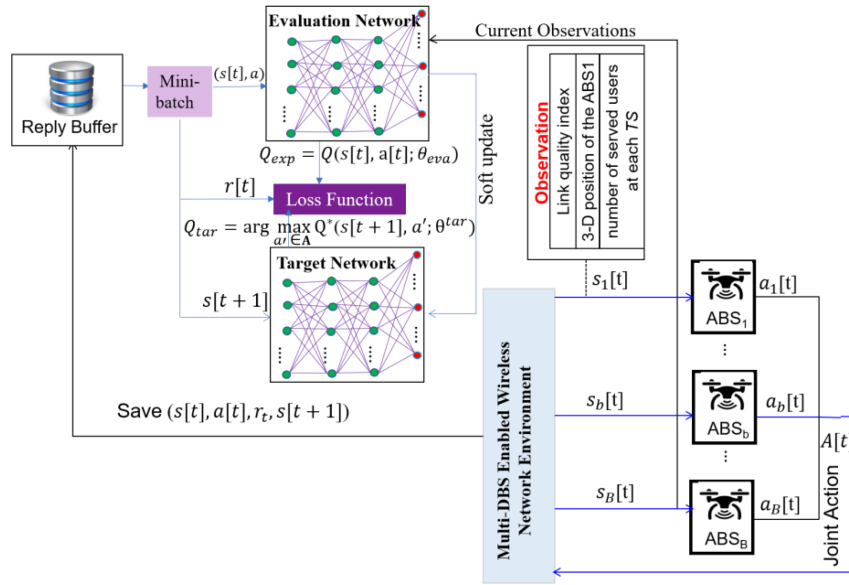


FIGURE 2. Overall architecture of the centralized multi-agent DRL-based MATC system.

epochs, number of LSTM layers, batch size, activation functions, and optimizers, were fine-tuned during training until the optimal model was achieved. In addition, dropout regularization was applied within the LSTM layer to prevent overfitting and ensure generalizability during training. The well-trained channel estimation model was subsequently uploaded to each DRL agent, enabling continuous control of the ABS trajectory based on the acquired link quality information and other environmental information.

Afterward, the training process of the proposed centralized multi-agent DRL framework is illustrated in Fig. 2. In addressing the MATC problem, each ABS is conceptualized as a DRL agent, comprising an online (policy) network, a target network, and an experience replay buffer. The online network plays a pivotal role in the DRL agent by facilitating the learning and evaluation of MATC policy performance. Network parameters are updated iteratively during the learning process, involving the estimation of Q -values to make precise decisions based on the observed system state at each iteration.

However, updating parameters may introduce fluctuations or divergences in the correlation between the current state and the target value. To mitigate this challenge, techniques such as target networks and experience replay buffers are employed to enhance the convergence of learning. The target networks replicate the structure of the online network, maintaining a copy of the online network parameters with a soft update mechanism to prevent instability and inconsistencies in the learning process. Experience replay buffer stores past experiences generated during the learning process, preventing correlated training data and promoting data independence. The centralized multi-agent DRL framework utilizes a mini-batch approach to randomly select samples from the replay buffer for training deep neural networks. This strategy enhances sample efficiency and learning stability in the multi-agent DRL process.

Algorithm 1 encapsulates the proposed MATC scheme by integrating centralized training with decentralized execution. The initialization phase (line 1 of Algorithm 1) involves setting up the entire multi-ABS communication environment and initializing the MATC strategy's multi-agent DRL network parameters. At each time slot, individual agents observe their respective states, denoted as $s_b[t]$, from the target environment. Moreover, the user link quality state is computed from the received amplitude and phase signals at each time slot through the uploaded link quality estimation (LQE) model (line 4 of Algorithm 1). Subsequently, each agent transmits its private observations $s_b[t]$ to a central server, which aggregates and processes all state information. The centralized DRL algorithm is then trained on this server, generating a policy based on the collective actions taken by all agents in the system. The central network takes the joint observations from all agents and produces a Q -value for each potential joint action.

Throughout the training phase (lines 6–8 of Algorithm 1), agents employ an ε -greedy exploration strategy to determine their actions $a_b[t]$ based on the current environmental state and the output of the centralized DRL network. During the initial learning stages, the DRL agent lacks a complete understanding of the environment, leading to uncertainty in the Q -function estimate. Consequently, the agent engages in exploration to some extent, mitigating this uncertainty and ensuring the development of an optimal policy [33]. Expressed mathematically, the probability of the DRL agent's action selection can be formulated as follows:

$$\hat{a}_b[t] = \begin{cases} \arg \max_{a_b[t] \in A} Q(s_b[t], a_b[t], \theta_b^{eva}) & \text{with probability } 1 - \varepsilon \\ \text{random } a_b[t] \in A & \text{with probability } \varepsilon, \end{cases} \quad (5)$$

Algorithm 1: Proposed MATC Strategy at the Training Stage.

Input: $MaxEpisode, \mathfrak{R}_s, A, M, \gamma, \beta, \varepsilon$;
Output: Learned 3D dynamic ABS trajectory strategy with maximum $r[t]$;

- 1: **Initialize** an experience replay buffer $\mathfrak{R}_{buf}$ the evaluation network $Q(\cdot)$, and the target network $Q^*(\cdot)$ with network parameters θ_{eva} and $\theta_{tar} = \theta_{eva}$, respectively, locations of all ABSs;
- 2: **for** $episode = 1, MaxEpisode$ **do**
- 3: Initialize the environment and get a system state;
- 4: Upload the LQE model to each ABS agent to estimate the user's link quality $m_{b,u}$;
- 5: **for** time step $t = 1, T$ **do**
- 6: **for** ABS $b = 1, \dots, B$ **do**
- 7: The central server determines an action $a_b[t]$ according to ε -greedy policy based on the observed $s_b[t]$ (t);
- 8: Takes the actions $a_b[t]$, gets the corresponding reward $r[t]$, according to (4) and moves to the next system state $s_b[t + 1]$;
- 9: **if** $experience < \mathfrak{R}_s$ **then**
- 10: The central server stores the ABS experiences in $\mathfrak{R}_{buf}$;
- 11: **else**
- 12: Overwrite the first saved experience with $e_b[t]$ in $\mathfrak{R}_{buf}$;
- 13: Sample a mini-batch of $\mathfrak{R}_{buf}$ historical experiences from M ;
- 14: Update the evaluation network parameters based on (6);
- 15: **if** $t \bmod \tau == 0$
- 16: Update the target network parameters, $\theta_{tar} = \theta_{eva}$;
- 17: **end if**
- 18: **end if**
- 19: **end for**
- 20: **end for**
- 21: **end for**

where the ε parameter represents the exploration rate. In practical implementation, the ε parameter is often initialized with a high value, close to unity, during the initial stages of the learning process. Over time, as the agent accumulates experience and gains confidence in its policy, ε is systematically reduced. Consequently, there is a high likelihood $1 - \varepsilon$ that the agent opts for an action from the Q -table, selecting the one with the maximum Q -value, $\hat{a}_b[t] = \arg \max_{a_b[t] \in A} Q(s_b[t], a_b[t], \theta_b^{eva})$. When the exploration probability ε is in effect, the agent randomly selects an action. In the event that the subsequent location extends beyond the target environment or breaches constraint $c3$, the ABS remains stationed at its current location. Moreover, the system incurs a penalty, denoted as p_3 , as elucidated in (4).

Algorithm 2: Online Application of the Proposed MATC Strategy.

Input: Trained network Q ; system state $s_b[t]$.
Output: Maximize $C[t]$ and $\Pi[t]$.

- 1: **for** ABS $b = 1, \dots, B$ **do**
- 2: Load the well-trained network Q from **Algorithm 1**;
- 3: **end for**
- 4: **for** $t = 1$ to T **do**
- 5: **for** ABS $b = 1, \dots, B$ **do**
- 6: ABS b selects an action $a_b[t]$ based on the observed system state $s_b[t]$ with the maximum Q -value of the trained Q -network;
- 7: update the system state ($s_b[t] = s_b[t + 1]$);
- 8: **end for**
- 9: $t = t + 1$;
- 10: **end for**

This penalty mechanism serves to discourage actions that would lead to the ABS being positioned beyond the designated target area or violating the specified constraint $c3$.

Upon executing the action $a_b[t]$, the reward $r[t]$ and the next system state $s_b[t + 1]$ are obtained. Meanwhile, the experiences $e_b[t] = (s_b[t], a_b[t], r[t], s_b[t + 1])$ are stored in a memory buffer $\mathfrak{R}_{buf}$ with a specified capacity (line 10 of Algorithm 1). After the ABS experience is stored to a capacity $\mathfrak{R}_s$, the DRL agent samples a mini-batch of size M historical experiences from $\mathfrak{R}_{buf}$ rather than training on a single ABS observation (lines 12–17 of Algorithm 1). Subsequently, the parameters of the deep Q -network (DQN) are optimized by minimizing the loss function $J(\theta[t])$ using the stochastic gradient descent method (line 14).

$$J(\theta[t]) = E [(y_{tar} - Q(s_b[t], a_b[t], \theta_{eva}))^2] \quad (6)$$

where $y_{tar} = r[s[t], a[t]] + \gamma \arg \max_{a'[t] \in A} Q^*(s[t], a'[t], \theta_{tar})$. Note that $\gamma \in (0, 1)$ is a discount factor for weighting future rewards, θ_{tar} is the target network parameter copied from the evaluation network parameter θ_{eva} (i.e., $\theta_{tar} = \theta_{eva}$) in every τ training iteration of the DQN to keep the training stable, which smooths out fluctuations (lines 15–17).

Online MATC Strategy Application: Once the training process is completed, the proficient ABS trajectory control policy can be implemented online. In the application stage, the ABSs continually observe the system state at each time slot. The ABS, utilizing the DNN, obtains the Q -values corresponding to all possible joint actions for the given system state. Subsequently, the action with the highest Q -value is selected. Algorithm 2 clarifies the real-time decision-making process employed by the ABS for trajectory control, aiming to maximize communication coverage and average network throughput over time.

Centralized multi-agent DRL-based MATC's computational complexity can be calculated by considering the complexity of the DNN structure used in training ABS agents. Typically, the computational complexity of a centralized multi-agent DRL model is highly dependent on the number of operations of the network model, including the neural network parameters (i.e., the input dimensions, the number of DNN layers, and the number of neurons in each hidden layer) as well as the number of agents involved. First, we assume that the DNN network of each agent contains L fully connected layers, and the f -th layer of the DNN network has n_f neurons. The time complexity of Algorithm 1 can be calculated as $O_{com}(B \cdot e \cdot T \cdot \sum_{f=0}^{F-1} n_f n_{f+1})$, where B is the number of DRL agents in the system, e denotes the number of episodes, and T is the number of iterations taken by the agents within one episode to get convergence. The steps include observing the current state, selecting an action, executing the action, receiving rewards, and transitioning to a new system state. The steps include observing the current state, selecting an action, executing the action, receiving rewards, and transitioning to a new system state. In our proposed system, we set the maximum value of episodes (e) and execution steps (T) via the trial-and-error method [39]. Above all, the proposed MATC strategy is suitable, universal, and effective for optimizing communication coverage and network throughput for ABS placement problems in ABS-aided wireless networks.

IV. SYSTEM EVALUATION

This section provides an overview of the implementation details and evaluation metrics discussed in Section IV-A. Subsequently, the empirical effectiveness of the proposed MATC strategy is validated through numerical results, aiming to substantiate its efficacy.

A. IMPLEMENTATION DETAILS

The experiments were conducted on a PC featuring a NVIDIA GeForce GTX 1080 Ti GPU, utilizing the Python programming language with the Keras framework on the front end and the TensorFlow framework on the back end. We trained an LSTM-based link quality estimation model using user channel data collected in the real-world environment. During the experiments, 4.92 million user channel samples were collected. This proposed LQE model is trained by inputting the preprocessed channel information to the LSTM algorithm, and the corresponding link quality level, i.e., the MCS index, is employed as a target value. Through the trial-and-error method [39], LSTM network hyperparameters, such as the number of epochs, number of LSTM layers, batch size, activation functions, and optimizers, are adjusted during training, while the LSTM network is trained until it is optimized. Therefore, the proposed LQE model employed five LSTM layers, each with 800 neurons and two fully connected (FC) layers, 1500 batch sizes, and 500 epochs. A dropout regularization technique is also used during training to prevent overfitting. We also train the model with Adamax optimizer, which has a learning rate of $2e - 3$. Then, the well-trained LQE model is uploaded to

TABLE 2. Network Parameters of the LSTM Algorithm

Hyperparameters	Value
LSTM layers	5 (each with 800 neurons)
Activation function	Tanh
Dropout rate	0.2
Batch size	1500
Optimizer (learning rate)	Adamax ($2e - 3$)
Epoch/Iteration	500
Loss	MSE

TABLE 3. Parameters of the Proposed DRL Algorithm

Hyperparameters	Value
No. of layers	2 [400, 300 neurons]
Optimizer (learning rate)	Adam ($\beta = 1e - 3$)
Activation function	ReLU, Softmax (last layer)
Discount factor γ , Mini-batch size M	0.95, 256
Target network update frequency τ	350
MCS index threshold (m_{min})	10
Initial and final exploration coefficient ϵ	$\epsilon_i = 0.99$, $\epsilon_f = 0.01$
Experience replay buffer size	800000
Max. training episodes, steps per episode	(1000, 120)

each DRL agent to continuously control the drone trajectory. Table 2 presents the details of the LSTM algorithm.

To estimate Q -values using deep Q -learning (DQL) with a DNN, we employed a fully connected feed-forward network consisting of two hidden layers with [400, 300] neurons. The Adam optimizer with a learning rate of $1e-3$ was utilized to enhance the network's performance. A rectified linear unit (ReLU) nonlinear activation function was applied in the hidden layers of the DNN to map state-action pairs into their respective Q -values. In the final output layer, a Softmax activation function was used to provide action feedback probabilities. A centralized multi-agent DRL algorithm has an output dimension equal to the number of joint discretized ABS flying directions.

The model training involved 1000 episodes, each comprising 120 execution steps. A replay buffer of size ($\mathfrak{R}_s$) 80000 was used, with a mini-batch size of 256. The target Q -network was updated every $\tau = 350$ steps. The initial exploration probability was set to 0.99, gradually decreasing to a final value of 0.01 to explore the environment effectively. The discount factor γ was set to 0.99 in our implementation setting. Detailed parameters of the algorithm are presented in Table 3.

The proposed MATC system was evaluated using two metrics: *i)* average system data rate ($\Pi[t]$), which is the average user data rate at the end of the testing period and (2) can be used to determine the average user data rate; *ii)* average Communication Coverage Score ($C[t]$) - which is the average communication coverage score at the end of the testing period. The coverage score of the proposed DMC strategy can be obtained from (1).

B. RESULTS AND ANALYSIS

As detailed in Section II-A, we constructed the proposed channel estimation model using real-world user channel data,

TABLE 4. Performance Comparisons Between Proposed LQE Method and Prior Work Using the Same User Channel Data

Learning Algorithm	Accuracy (%)
CNN [7]	93.52
<i>k</i> -means [8]	85.39
Random Forest [29]	89.17
Time-Series CNN [28]	97.92
Proposed LSTM	98.94

opting for practical data over theoretical channel modeling [9], [10], [11], [12], [13], [14], [15], [16], [24], [25], [26], [27]. As discussed in Section III-B, we collected 3.94 million samples of user channels to build and validate the effectiveness of the proposed LQE method for the MATC policy. In particular, we trained an LSTM-based LQE model using real-world user channel data. The well-trained LQE model can continuously assess the link quality level of ground users at each time slot. Experimental results demonstrated that the LSTM-based LQE models achieved a remarkable 98.94% accuracy in estimating a user’s link quality level, as indicated in Table 4.

The proposed system leverages both amplitude and phase signal information to extract discriminative propagation characteristics from dynamic environments and time-varying channel conditions. To further assess the performance of the proposed method, we conducted additional experiments comparing it with prior approaches such as CNN [7], *k*-means [8], random forest [29], and time-series CNN [28]. It is important to note that the comparison involves different environments, user channel data, and model assumptions. To enhance the representativeness of the comparison, we utilize the proposed user’s channel data to evaluate the proposed link quality estimation model against the aforementioned prior works [7], [8], [28], [29], as presented in Table 4.

Experiments showed that the learning method introduced in [8] exhibits poor scalability due to large datasets, leading to significantly lower performance (85.39%). Although CNN [7] and time-series CNN [29] classifiers achieve higher scores than the method in [8], the LSTM-based LQE model proposed here is superior to others for several reasons. Firstly, the LSTM algorithm effectively determines link quality over time by mitigating issues such as vanishing and exploding gradients through gate mechanisms (i.e., an input gate, a forget gate, and an output gate). Conventional learning approaches [8], [29] struggle with large-scale user channel data in dynamic wireless environments due to limited adaptability, high gradient disappearance, and low learning capacity. Second, our proposed system excels in extracting distinctive air-to-ground propagation characteristics based on the received amplitude and phase signals.

Fig. 3 illustrates the average communication coverage scores achieved by the proposed method compared with the baseline methods. To validate the effectiveness of our proposed MATC strategy, we compared its performance to the distributed multi-agent DRL method [32] and the centralized multi-agent *Q*-learning baseline method [24], [25], [26]. In the distributed implementation, each ABS trains its own model

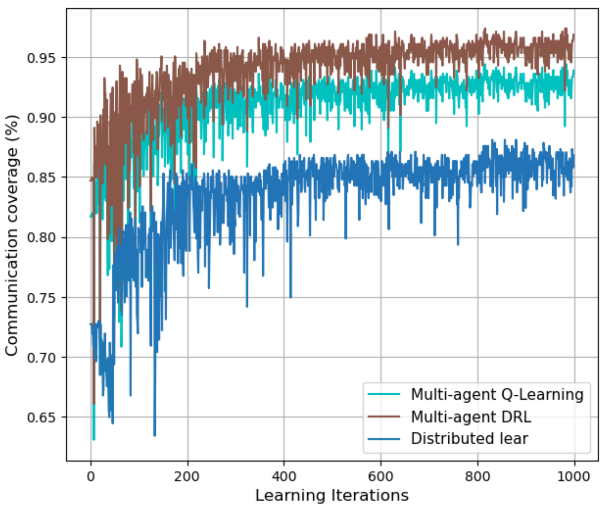


FIGURE 3. Average communication coverage performance comparison.

based on the observed state information without sharing network parameters with other ABSs or the central server. The second baseline algorithm is known as centralized multi-agent *Q*-learning, where all ABSs send their state information to the central server in each episode to optimize the global model. With increasing learning iterations, all learning approaches show increasing average communication coverage. In the first 150 episodes, all learning algorithms tend to achieve lower coverage performance. This is because the system does not thoroughly train the neural networks; instead, it tries to get enough experience in the replay buffer by randomly explore the unknown environment to optimize the learning policy. With increasing training episodes, learning agents develop an optimal MATC scheme by gaining a deeper understanding of the environment.

The proposed MATC strategy achieves a 97.41% communication coverage score while maintaining satisfactory network throughput for ground users. In contrast, the centralized multi-agent *Q*-learning approach and the distributed multi-agent DRL scheme have communication coverage scores of 94.09% and 88.11%, respectively. The MATC strategy improved coverage performance by 9.3% and 3.32% compared to distributed multi-agent DRL and centralized multi-agent *Q*-learning, respectively. However, it significantly increased the average user throughput, as shown in Figs. 4 and 5. Modeling accurate ABS trajectories using the multi-agent *Q*-learning approach becomes more challenging as the network size increases due to curse dimensionality problems. Besides, decentralized learning methods can suffer from reduced accuracy and suboptimal performance due to the limited information available to each agent. Notably, the proposed method consistently outperforms the baseline centralized multi-agent *Q*-learning and distributed multi-agent DRL methods, providing a higher communication coverage ratio.

In addition, the effects of the number of neurons in a fully connected layer of a DNN and the discount factor on the average communication coverage are investigated and

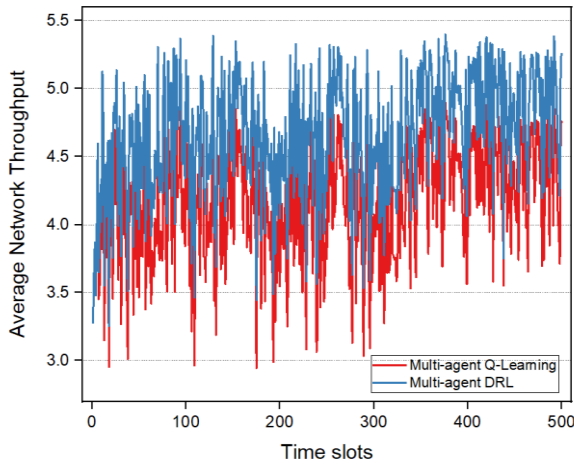


FIGURE 4. Average user throughput performance of centralized multi-agent DRL with centralized multi-agent Q-learning.

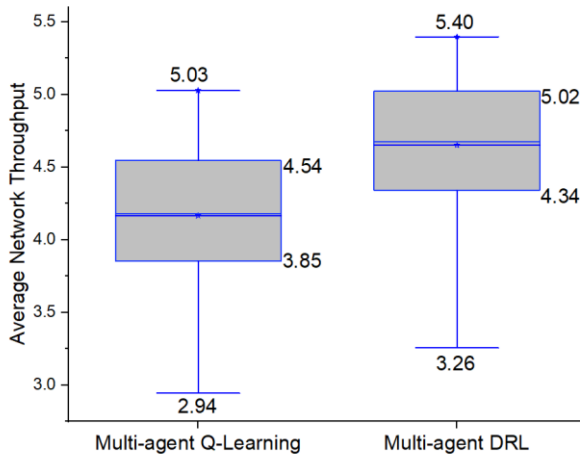


FIGURE 5. The distribution of average user throughput performance in different learning approaches within 500 time slots.

TABLE 5. MATC Performance With Different DRL Network Hyperparameters

Network model	Coverage (%)		
	Discount Factor γ		
	0.8	0.9	0.95
Model-I	90.35	91.97	93.19
Model-II	92.89	93.88	97.41
Model-III	93.04	95.21	96.38

presented in Table 5. Three models are considered: Model-I with 250 and 300 neurons, Model-II with 350 and 400 neurons, and Model-III with 500 and 600 neurons in the first and second layers, respectively. All models underwent testing with 1000 training episodes, and the average communication coverage performance is reported in Table 5.

Consistent with experiments using the same network structure, it is observed that the average communication coverage increases with higher discount factors (γ). Specifically, Model II and Model III demonstrate improvements in average coverage scores by 3.42% and 2.73%, respectively, when γ changes

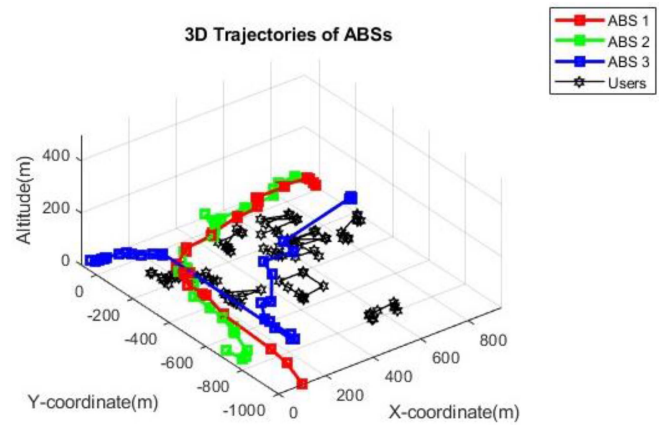


FIGURE 6. ABS trajectories in different time slots.

from 0.8 to 0.99. Additionally, the system's performance increases as the fully connected layer contains more neurons. In selecting the proposed model (Model-II), a tradeoff is considered between system performance and algorithm efficiency.

Fig. 4 illustrates the average network throughput of the proposed method and multi-agent Q-learning methods over different time slots. The graph reveals that the average user throughput performance fluctuates over time in a dynamic wireless environment. This variability is attributed to the tradeoff between coverage and communication quality, causing ABS positions to become suboptimal. Moreover, the centralized multi-agent DRL algorithm's performance was compared with the multi-agent Q-learning algorithm's performance in [24], [25]. Conventional multi-agent Q-learning algorithm offers less average user network throughput than the proposed MATC system, as shown in Figs 4 and 5. Due to learning instability, estimation of the expected rewards using a multi-agent Q-learning approach has a high variance. As shown in Fig. 5, the majority of the user throughput performance (i.e., around 75% of the served users) occurred in the range of 3.85 bit/symbol to 4.54 bit/symbol in the multi-agent Q-learning approach, whereas 4.34 bit/symbol to 5.02 bit/symbol in the proposed MATC strategy. Based on the experiments, the proposed system demonstrated promising and reasonable communication performances from both a communication coverage and average user throughput perspective. Consequently, ABS locations are adjusted based on received user signals until the ground user throughput surpasses a predetermined threshold. The integration of aerial base stations and multi-agent DRL contributes to improved coverage scores and enhanced user throughput in wireless communications.

The ABS trajectories are shown in Fig. 6, where the red line represents ABS 1, the green line signifies ABS 2, the blue line denotes ABS 3, and the black line represents the movement trajectory of ground users (10 users in this simulation). ABSs continuously fly between 30 m and 50 m altitude in the target area. We observed that ABSs fly at lower altitudes when ground users are concentrated; while when ground users are

spread out across the target area, ABSs fly at higher altitudes to maintain communication links. Therefore, in our proposed MATC strategy, ABSs change their flight trajectories based on the received state information to maximize system rewards.

V. CONCLUSION AND FUTURE WORK

In this paper, we have presented an autonomous trajectory control method for multi-UAV-enabled wireless communications that maximizes communication coverage and network throughput. Instead of using conventional channel models, we first proposed an LSTM-based link quality estimation model to assess ground user link quality over time. Then, we utilized a multi-agent DRL to train the MATC scheme to continuously control network performance along the ABS trajectory. The proposed MATC system is implemented on experimental user channel data in a real-world environment. Numerical results showed that the proposed MATC strategy achieves promising and reasonable performance in terms of both communication coverage and user throughput. In future work, we will further consider the trajectory of ABSs constrained to 360 degrees of angle rather than discrete directions. Speed and other parameters will also be considered as optimization objectives. We also plan to propose a federated multi-agent deep reinforcement learning approach for ABS trajectory and resource allocation schemes to reduce learning complexity and protect ground user data privacy.

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GETANEH BERIE TAREKEGN (Member, IEEE) received the B.Sc. degree in computer science from the University of Gondar, Gondar, Ethiopia, in 2012, the M.Sc. degree in electrical engineering, and the Ph.D. degree in computer science from the National Taipei University of Technology, Taipei, Taiwan, in 2019 and 2022, respectively.

He is currently a Postdoctoral Research Fellow with the Department of Electrical and Computer Engineering and the Institute of Electrical and Control Engineering, National Yang Ming Chiao Tung University, Hsinchu, Taiwan. His research interests include wireless communications, UAV-based wireless communications, positioning and navigation technology, reconfigurable intelligent surfaces for wireless communications, Artificial Intelligence for 5G/B5G/6G wireless networks, and computer vision.



RONG-TERNG JUANG (Member, IEEE) received the M.S. and Ph.D. degrees in electrical engineering from the National Taipei University of Technology, Taipei, Taiwan, in 2002 and 2007, respectively. Since 2020, he has been an Associate Professor with the Institute of Aerospace and System Engineering, National Taipei University of Technology. His research interests include cell planning, mobile localization, and resource management in wireless cellular networks.



BELAYNEH ABEBE TESFAW received the B.Sc. and M.Sc. degrees in information technology from Debre Markos University, Debre Markos, Ethiopia, in 2017 and 2021, respectively. He is currently working toward the Ph.D. degree in electrical engineering and computer science with the National Taipei University of Technology, Taipei, Taiwan. His research interests include wireless communications, UAV-based wireless communications, reconfigurable intelligent surfaces for wireless communications, deep learning, and image processing.



HSIN-PIAO LIN (Senior Member, IEEE) received the B.S. degree in communication engineering from National Chiao-Tung University, Hsinchu, Taiwan, in 1986. He received the M.S. and Ph.D. degrees from the Electrical Engineering Research Laboratory with the University of Texas, Austin, TX, USA, in 1992 and 1997, respectively.

Since 2015, he has been the Director of the Multiple Antenna System Research and Development Center (MASRDC) with the National Taipei University of Technology, where he is currently the Founding Professor. His research interests include the areas of cognitive radio, Wi-Fi/LTE network communications, radio propagation modeling, smart antenna systems, and applications of machine/deep learning in wireless communications.



HUAN-CHIA HSU received the B.S. degree in electrical engineering from National Chung Cheng University, Chiayi, Taiwan, in 2022. He is currently working toward the M.S. degree in electrical engineering with National Yang Ming Chiao Tung University, Hsinchu, Taiwan. He has been a member of FAITHLAB in the Department of Electrical and Control Engineering, Hsinchu, since 2022. His research interests include in the fields of Internet of Things (IoT), deep learning, unmanned aerial vehicles (UAVs), and sensors.



ROBEL BERIE TAREKEGN received the B.Sc. degree in electrical and computer engineering from Wolaita Sodo University, Wolaita Sodo, Ethiopia, in 2019. He is currently working toward the M.Sc. degree in electrical engineering and computer science with National Yang Ming Chiao Tung University, Hsinchu, Taiwan. He has been a member of FAITHLAB in the Department of Electrical and Control Engineering, Hsinchu, since 2023. His research interests include Internet of Things (IoT), wireless communication, signal processing, and cognitive radio.



LI-CHIA TAI (Senior Member, IEEE) received the B.S. degree in physics from University of California, Los Angeles, Los Angeles, CA, USA, in 2014. He received the Ph.D. degree in electrical engineering and computer sciences from University of California Berkeley, Berkeley, CA, USA, in 2020. Since 2021, he has been an Assistant Professor and the founder of FAITHLAB with the Department of Electrical and Computer Engineering and the Institute of Electrical and Control Engineering, National Yang Ming Chiao Tung University (NYCU), Hsinchu, Taiwan. His research interests include wearable sensors, biomedical devices, optoelectronics, health analytics based on deep learning, and Internet of Things (IoT).